Low Light Image Enhancement using Homomorphic Filtering

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Problem and Related Work

Our Problem:

- Low-light images suffer from low visibility, noise and poor contrast
- Hard for humans and Computer Vision (Sound familiar professor?) to interpret
- Need a method that improves illumination without blowing out details

Related Work:

- Histogram Equalization - boosts contrast but amplifies noise
- Gamma Correction - simple brightness adjustment but weak on uneven lighting
- Retinex Based Models - good but computationally heavier
- Homomorphic filtering - classic approach using illumination-reflectance separation

Technical Approach Part 1

Illumination Reflectance Model:

An image is reflected as f(x,y) = i(x,y) * r(x,y). i is illumination and r is reflectance.

The steps:

1. Convert to log domain - turns multiplication into addition
2. Apply high-pass filter in frequency domain to suppress illumination
3. Boost reflectance components for better detail
4. Exponentiate and normalize

Technical Approach Part 2

Methods Compared:

- Homomorphic filter
- Histogram Equalization
- Gamma Correction

No-Reference Quality Metrics:

- NIQE
- BRISQE
- Entropy
- Tenengrad Sharpness

Project Demo Part 1

Pipeline Demonstration:

- Input: Raw ExDark Image
- Preprocessing: Convert to grayscale for some methods
- Run all enhancement algorithms
- Output

Project Demo Part 2

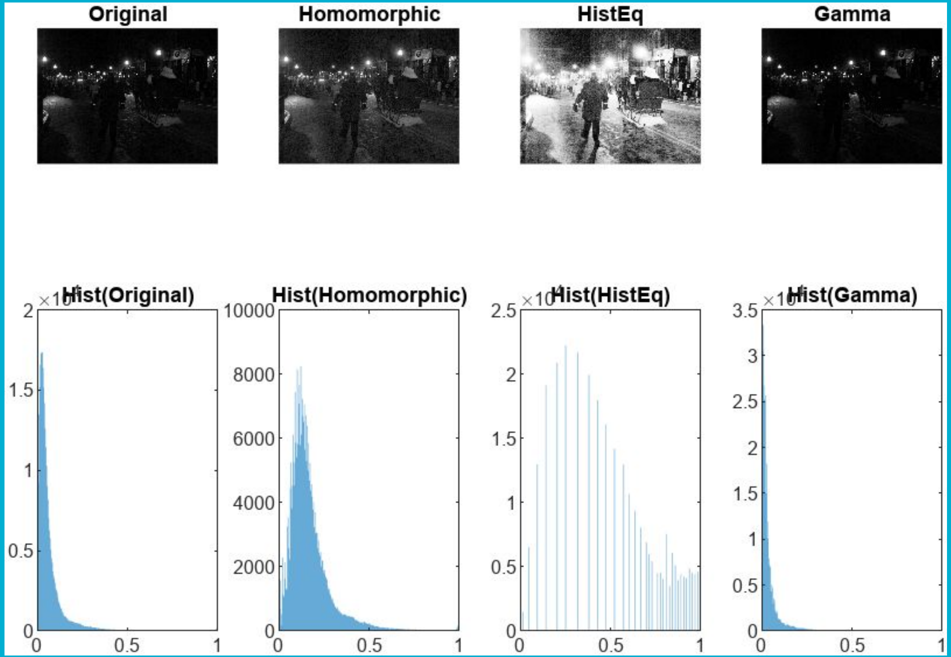
Code Flow Overview:

- Read Image - im2double()
- Apply homomorphic_enhance()
- Apply baselines
- Compute all the No-Reference Quality Metrics
- Save Images
- Generate Figures

Results Part 1

Visual Comparison:

- Homomorphic produces smoother brightness and less noise
- HistEq produces high noise and over contrast
- Gamma brightens it but can't handle uneven illumination



Results Part 2

Metric Comparison:

- Table: All the No-Reference Quality Metrics
- What Method Scores best
- Homomorphic gave us the best balance of everything

```
Homomorphic:
     3.7000    25.9159     6.1747     0.0018
HistEq:
     4.1543    22.1065     4.5759     0.0023
Gamma       :
     4.0077    30.3352     4.7771     0.0014
```

Metric	Low Value	High Value	Note
NIQE	Natural	Unnatural	More sensitive to noise and unnatural brightness patterns
BRISQUE	Natural, good quality	Distorted, noisy, unnatural	Strong at detecting over-processin g
Entropy	flat, low-contrast , overly smoothed, or very dark images	better contrast, richer details, more visible structure	
Tenengrad	Blurry or low-detail image	Sharp, strong edges	

Conclusion

Key Takeaways:

- Homomorphic Filtering is effective best for low-light enhancement
- It outperformed HistEq and Gamma
- No-reference Metrics confirm improved quality
- Parameter Tuning is critical for success
  - Optimal parameters were found to be:
  - Gamma L = 0.6, Gamma H = .5, Do = 250, n = 1